TAK Server Public Facing Domain

Advanced: This needs to be done on a real server

Manage the DNS

- Go to the server service you use → my products → your domain → manage DNS
  - Select “Add New Records”
  - Type: A
  - Name: choose a name
  - IP Address: the IP address of your server
  - Select save

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- The following steps are for the use of the downloaded scripts:
  - https://drive.google.com/drive/folders/1y3nhDPP0UgbdtfyUmYyTP0slfSGdxiol

takserver_createLECerts.sh	8/12/2024 3:32 PM	Shell Script	4 KB
takserver_renewLECerts.sh	8/12/2024 3:32 PM	Shell Script	2 KB

- SSH into you TAK Server
- If the scripts aren't downloaded, download them to the base directory
  - **cd ~**
  - From the local system open a terminal to the files download location
    - **scp <file> <username>@<IP address>:~/**
- Que the LetsEncrypt Script
  - **sudo chmod +x takserver_createLECerts.sh**
- Enter the password for the atak user if asked
  - **atakatak**
- Execute script
  - **sudo ./takserver_createLECerts.sh**

Add the DNS to your Server

- From the command line:
  - Prompted to enter a email so it can get ahold of you: <your email>
  - Agree to the terms of service: y
  - Sharing of email choose: y or n
  - Enter your domain name: <name for the record>.<your domain>.<net,com,org,.etc>
  - Enter certificate name: <usually good to keep it the same as the domain name.
  - If done right you will see the following:

```
Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/swat1.taktical.net/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/swat1.taktical.net/privkey.pem
This certificate expires on 2024-06-29.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.
We were unable to subscribe you the EFF mailing list because your e-mail address appears to be invalid. You can try again later by visiting
https://act.eff.org.
```

TAK Server Public Facing Domain Setup - CL

- To create a public facing domain for the TAK Server using the command line, run the following:
  - Install snapd
    - **sudo yum install snapd -y**
  - Enable snapd
    - **sudo systemctl enable --now snapd.socket**
  - Create the symbolic link for snap
    - **sudo ln -s /var/lib/snapd/snap /snap**
  - Configure the firewall by adding the necessary port
    - **sudo firewall-cmd --zone=public --permanent --add-port=80/tcp**
  - Reload the firewall to accept the changes
    - **sudo firewall-cmd --reload**
  - Verify the firewall changes
    - **sudo firewall-cmd --list-all**

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- Restart snapd and wait for the system to load
  - **sudo systemctl restart snapd.seeded.service**
  - **snap wait system seed.loaded**
- Using snapd install certbot
  - **sudo snap install --classic certbot**
- Create the symbolic links to certbot
  - **sudo ln -s /snap/bin/certbot /usr/bin/certbot**
- Begin our certificate signing process for the server
  - **sudo certbot certonly --standalone**

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- Fill in the following information as it pops up
  - Email address: <your email address or c to cancel>
  - Terms of Service: yes
  - Share email: yes or no
  - Enter domain name: ds.hunthideaway.net → <name>.<domain>.<com,net,org,…>
    - <domain name you setup for the domain you own, if you’d like to purchase one you can. This should’ve been setup in slide 2>
- Verify the new singed certificate by viewing its contents
  - **read -p 'Certificate Name (FQDN) [ex: tak.domain.com]: ' certNameVar**
  - **openssl x509 -text -in /etc/letsencrypt/live/$certNameVar/fullchain.pem -noout**

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- Conduct a certificate renewal dry run
  - **sudo certbot renew --dry-run**
- Create our PKCS12 certificate from our signed cert and private key
  - **sudo openssl pkcs12 -export -in /etc/letsencrypt/live/$certNameVar/fullchain.pem -inkey /etc/letsencrypt/live/$certNameVar/privkey.pem -out takserver-le.p12 -name $certNameVar -password pass:atakatak**
  - You can view the contents
    - **sudo openssl pkcs12 -info -in takserver-le.p12**
- Create our Java Keystore from our PKCS12 cert
  - **sudo keytool -importkeystore -srcstorepass atakatak -deststorepass atakatak -destkeystore takserver-le.jks -srckeystore takserver-le.p12 -srcstoretype pkcs12**

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- Move the certificate to the TAK Cert directory
  - **sudo mv takserver-le.jks /opt/tak/certs/files**
- Restore our permissions to default
  - **sudo chown -R tak:tak /opt/tak**
  - **cd /opt/tak**
  - **sudo systemctl stop takserver**
- Replace the existing 8446 connector
  - **sed -i 's|<connector port="8446" clientAuth="false" _name="cert_https"/>|<connector port="8446" clientAuth="false" _name="LetsEncrypt" keystore="JKS" keystoreFile="certs/files/takserver-le.jks" keystorePass="atakatak"/>|g' /opt/tak/CoreConfig.xml**

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- Modify the CoreConfig files
  - **rm /opt/tak/CoreConfig.example.xml**
  - **cp /opt/tak/CoreConfig.xml /opt/tak/CoreConfig.example.xml**
- Start the server
  - **sudo systemctl start takserver**

- Note: Doing it this way you will need to manually renew the server monthly. Renewing the server will be covered in the next slide.

TAK Server Public Facing Domain Setup - CL

- Renewal:
  - Conduct a certificate renewal dry run to verify permissions and path
    - **sudo certbot renew --force-renewal**
  - Create our PKCS12 certificate from our signed certificate and private key
    - **sudo openssl pkcs12 -export -in /etc/letsencrypt/live/$certNameVar/fullchain.pem -inkey /etc/letsencrypt/live/$certNameVar/privkey.pem -out takserver-le.p12 -name $certNameVar -password pass:atakatak**
    - You can view the PKCS12 content
      - sudo openssl pkcs12 -info -in takserver-le.p12
  - Create our Java Keystore from our PKCS12 certificate
    - **sudo keytool -importkeystore -srcstorepass atakatak -deststorepass atakatak -destkeystore takserver-le.jks -srckeystore takserver-le.p12 -srcstoretype pkcs12**

TAK Server Public Facing Domain Setup - CL

- Remove the old jks and p12 files
  - **sudo rm /opt/tak/certs/files/takserver-le.jks**
  - **sudo rm /opt/tak/certs/files/takserver-le.p12**
- Move the certificate to the TAK certificate directory
  - **sudo mv takserver-le.jks /opt/tak/certs/files/**
  - **sudo mv takserver-le.p12 /opt/tak/certs/files/**
- Restore our permissions to default
  - **sudo chown -R tak:tak /opt/tak**
- Restart TAK Server
  - **sudo systemctl stop takserver**
  - **sudo systemctl start takserver**

Connecting with TAK Devices

- To connect to the TAK Server with your TAK devices using the domain name:
  - Navigate to the settings → network preferences → Tak Servers
  - Select the three dots in the top left of the screen and select add.
  - Name: Enter a display name for the server.
  - Address: enter in the public facing domain name <address>.<domain>.<com,org,net,etc.>
  - Enable: Enroll for client certificate
    - The rest should be disabled.
  - Select OK
  - You should be prompted to enter the credentials for the server.
    - Note: the credentials are created in the server dashboard